

2020 GI

呂志得 (Esophageal Ca)

1. 一個病人吞嚥困難請問你怎麼 approach

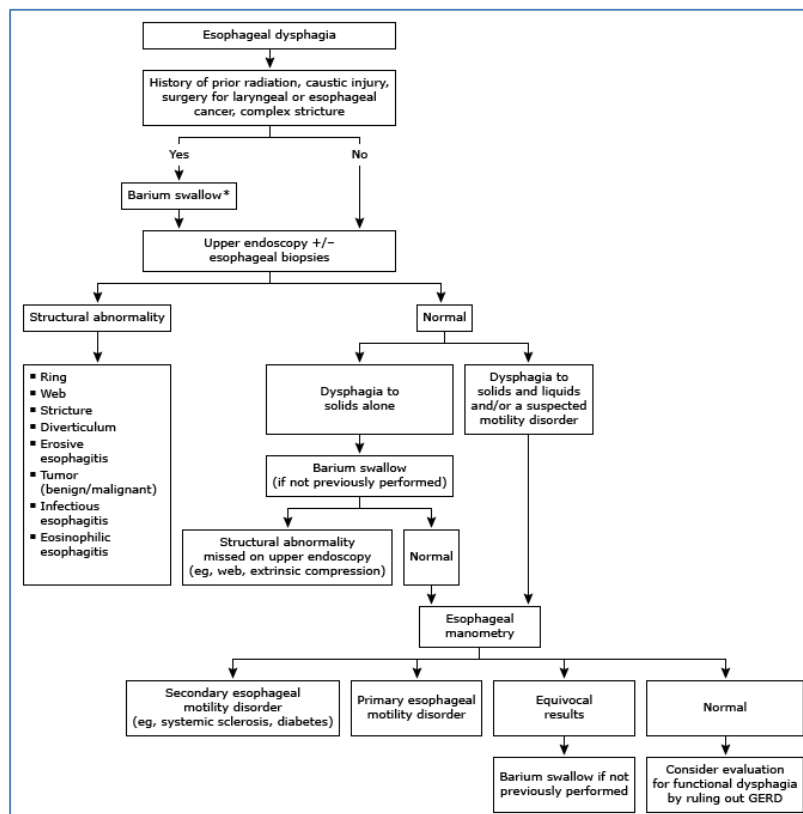
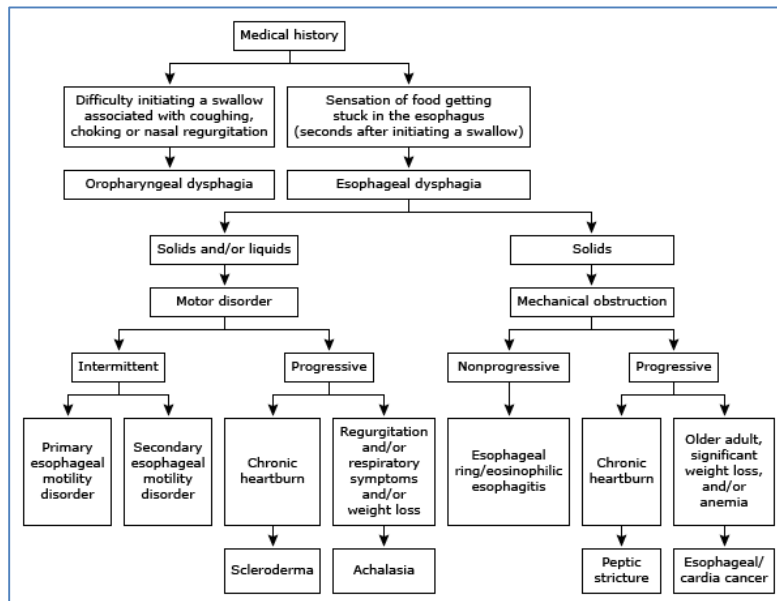
Ans:

(1) Dysphagia 先區分 oropharyngeal dysphagia/esophageal dysphagia

→若為 esophageal dysphagia →由 solids+liquids/solids dysphagia 區分

Esophageal neoplasm dysphagia:

Sensation of food getting stuck in the esophagus(seconds after initiating a swallow) 、Solids 、Progressive



(2) Associated risk factor?

Alcohol, tobacco, achalasia, obesity, GERD, Barrett's esophagus

(3) Workup exams (Ref. NCCN version 2.2021)

Upper GI endoscopy and biopsy; Chest/abdominal CT with oral and IV contrast; Pelvic CT with contrast as clinically indicated; FDG-PET/CT (skull base to mid-thigh) if no evidence of M1 disease; CBC and comprehensive chemistry profile; Endoscopic ultrasound (EUS), if no evidence of M1 unresectable disease

2. 證實有 middle third 食道癌,請問怎麼 approach

Ans:

(1) Adenocarcinoma/SqCC ?

(2) Staging: (Ref. NCCN version 2.2021)

- Chest/abdominal CT with oral and IV contrast
- FDG-PET/CT
- Endoscopic ultrasound (EUS)
- Endoscopic resection (ER) is essential for the accurate staging of early-stage cancers (T1a or T1b)
- Early stage cancers can best be diagnosed by ER
- Biopsy of metastatic disease as clinically indicated
- MSI by PCR/MMR by IHC, and PD-L1 testing if metastatic disease is documented/suspected
- HER2 testing if metastatic adenocarcinoma is documented/suspected
- If sufficient tissue is available after the above testing has been completed, next-generation sequencing (NGS) may be considered
- Bronchoscopy, if tumor is at or above the carina with no evidence of M1 disease
- Assign Siewert category
- Nutritional assessment and counseling
- Smoking cessation advice, counseling, and pharmacotherapy as indicated
- Screen for family history

3. Neoadjuvant CCRT 你們怎麼做

Ans:

(1) 各醫院作法不同，以下為本醫院作法之一供參考

- Concurrent CCRT
- R/T:
 - GTV: GTV of the primary (GTVp) + GTV of lymph nodes (GTVn)
 - CTV1: [GTVp + 5-cm craniocaudal & 1-cm radial margin along the esophagus] + [GTVn + 1-cm margin]
 - CTV2: [GTVp + 2-cm craniocaudal and 1-cm radial margin along the esophagus] + [GTVn + 1-cm margin]
 - PTV: CTV +1-cm in all directions
 - PTV1、2 sequentially treated to 36、41.4–50.4 Gy

- C/T:

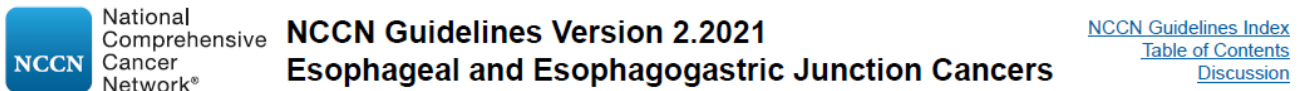
- Cisplatin (20 mg/m² daily, D1–4) + 5FU (800 mg/m² daily, D1–4) Q4W

(2) 其他 Ref: Target Volume Delineation for Conformal and Intensity-Modulated Radiation Therapy

Table 1 Summary of guidelines for contouring esophageal cancer

Tumor location	Definition	GTV to CTV margin	CTV to PTV margin	Elective nodal coverage	Dose
Upper esophagus	Above carina	Primary: 3~5 cm longitudinally; 0.5~1 cm circumferentially Involved lymph node: 0.5~1 cm in all directions	No IGRT: 0.5~1.0 cm IGRT: 0.5 cm	Periesophageal, mediastinal, supraclavicular	Neoadjuvant: 40~50.4 Gy in 1.8~2 Gy per fraction Definitive: 50.4 Gy (up to 66 Gy for tumor at cervical esophagus) in 1.8~2 Gy per fraction
Lower esophagus	Below carina	Same as upper esophagus	Same as upper esophagus	Periesophageal, mediastinal, perigastric, celiac	Neoadjuvant: 40~50.4 Gy in 1.8~2 Gy per fraction Definitive: 50.4 Gy in 1.8~2 Gy per fraction

(3) 其他 Chemotherapy Ref: NCCN Version 2.2021 (#編號同 Slide 內 Ref)



PRINCIPLES OF SYSTEMIC THERAPY—REGIMENS AND DOSING SCHEDULES^K

PREOPERATIVE CHEMORADIATION

PREFERRED REGIMENS

Paclitaxel and carboplatin
Paclitaxel 50 mg/m² IV on Day 1
Carboplatin AUC 2 IV on Day 1
Weekly for 5 weeks¹

Fluorouracil^b and oxaliplatin

Oxaliplatin 85 mg/m² IV on Day 1
Leucovorin 400 mg/m² on Day 1
Fluorouracil 400 mg/m² IV Push on Day 1
Fluorouracil 800 mg/m² IV continuous infusion over 24 hours daily on Days 1 and 2
Cycled every 14 days for 3 cycles with radiation^{2,1}

Fluorouracil 300 mg/m² IV continuous infusion over 24 hours daily for 4 days (over 96 hours) weekly
Oxaliplatin 85 mg/m² IV over 2 hours on Day 1
Cycled every 14 days for 3 cycles with radiation⁶⁵

Capecitabine and oxaliplatin

Oxaliplatin 85 mg/m² IV on Days 1, 15, and 29 for 3 doses
Capecitabine 625 mg/m² PO BID on Days 1–5 weekly for 5 weeks⁶⁶

OTHER RECOMMENDED REGIMENS

Fluorouracil and cisplatin

Cisplatin 75–100 mg/m² IV on Days 1 and 29
Fluorouracil 750–1000 mg/m² IV continuous infusion over 24 hours daily on Days 1–4 and 29–32
35-day cycle⁴

Cisplatin 15 mg/m² IV daily on Days 1–5
Fluorouracil 800 mg/m² IV continuous infusion over 24 hours daily on Days 1–5
Cycled every 21 days for 2 cycles⁵

Capecitabine and cisplatin

Cisplatin 30 mg/m² IV on Day 1
Capecitabine 800 mg/m² PO BID on Days 1–5
Weekly for 5 weeks⁶⁷

Irinotecan and cisplatin

Irinotecan 65 mg/m² IV on Days 1, 8, 22, and 29
Cisplatin 30 mg/m² IV on Days 1, 8, 22, and 29⁶

OTHER RECOMMENDED REGIMENS—continued

Paclitaxel and fluoropyrimidine

Paclitaxel 45–50 mg/m² IV on Day 1 weekly
Fluorouracil 300 mg/m² IV continuous infusion daily on Days 1–5
Weekly for 5 weeks⁷

Paclitaxel 45–50 mg/m² IV on Day 1
Capecitabine 625–825 mg/m² PO BID on Days 1–5
Weekly for 5 weeks⁷

#1: Paclitaxel + carboplatin (Ref: 2012 NEJM CROSS Trial)

Weekly Paclitaxel 50 mg/m² IV on Day 1 + Carboplatin AUC 2 IV on Day 1, and total 5 weeks.

#2: Fluorouracil + oxaliplatin (Ref: 2014 Lancet Oncol ACCORD 17)

Biweekly oxaliplatin 85 mg/m² IV on Day 1 + Leucovorin 400 mg/m² on Day 1 + Fluorouracil 400 mg/m² IV Push on Day 1 + Fluorouracil 800 mg/m² IV continuous infusion over 24 hours daily on Days 1 and 2, and total 3 cycles with radiation.

#4: Fluorouracil + cisplatin (Ref: 2008 JCO CALGB 9781)

Cisplatin 75–100 mg/m² IV on Days 1 and 29 + Fluorouracil 750–1000 mg/m² IV continuous infusion over 24 hours daily on Days 1–4 and 29–32, and 35-day cycle.

#5: Fluorouracil + cisplatin (Ref: 2007 JCO FFCD 9102)

Triweekly Cisplatin 15 mg/m² IV daily on Days 1-5 + Fluorouracil 800 mg/m² IV continuous infusion over 24 hours daily on Days 1-5 , and cycled every 21 days for 2 cycles.

4. Definitive CCRT 你們怎麼做

Ans:

(1) 各醫院作法不同，以下為本醫院作法之一供參考（基本上做法和 neoadjuvant CCRT 相同，差別在 RT dose 提高）

- Concurrent CCRT
 - GTV: GTV of the primary (GTVp) + GTV of lymph nodes (GTVn)
 - CTV1: [GTVp + 5-cm craniocaudal & 1-cm radial margin along the esophagus] + [GTVn + 1-cm margin]
 - CTV2: [GTVp + 2-cm craniocaudal and 1-cm radial margin along the esophagus] + [GTVn + 1-cm margin]
 - PTV: CTV +1-cm in all directions
 - PTV1、2 sequentially treated to 36、50–50.4 Gy
 - GTV with the relevant PTV was boosted up to 60–66.6 Gy if dose constraints of the organs at risk could be met.
- C/T
 - Cisplatin (25 mg/m²) + 5FU (1000 mg/m²) QW
 - Cisplatin (20 mg/m² daily, D1–4) + 5FU (800 mg/m² daily, D1–4) Q4W

(4) 其他 Chemotherapy Ref: NCCN Version 2.2021 (#編號同 Slide 內 Ref)

 National Comprehensive Cancer Network®	NCCN Guidelines Version 2.2021 Esophageal and Esophagogastric Junction Cancers	NCCN Guidelines Index Table of Contents Discussion
PRINCIPLES OF SYSTEMIC THERAPY—REGIMENS AND DOSING SCHEDULES^k		
DEFINITIVE CHEMORADIATION (NON-SURGICAL) PREFERRED REGIMENS Paclitaxel and carboplatin Paclitaxel 50 mg/m ² IV on Day 1 Carboplatin AUC 2 IV on Day 1 Weekly for 5 weeks ¹ Fluorouracil and oxaliplatin^b Oxaliplatin 85 mg/m ² IV on Days 1, 15, and 29 for 3 doses Fluorouracil 180 mg/m ² IV daily on Days 1–33 ³ Oxaliplatin 85 mg/m ² IV on Day 1 Leucovorin 400 mg/m ² IV on Day 1 Fluorouracil 400 mg/m ² IV Push on Day 1 Fluorouracil 800 mg/m ² IV continuous infusion over 24 hours daily on Days 1 and 2 Cycled every 14 days for 3 cycles with radiation followed by 3 cycles without radiation ² Capecitabine and oxaliplatin Oxaliplatin 85 mg/m ² IV on Days 1, 15, and 29 for 3 doses Capecitabine 625 mg/m ² PO BID on Days 1–5 weekly for 5 weeks ^{6,5} Fluorouracil and cisplatin Cisplatin 75–100 mg/m ² IV on Day 1 Fluorouracil 750–1000 mg/m ² IV continuous infusion over 24 hours daily on Days 1–4 Cycled every 28 days for 2 cycles with radiation followed by 2 cycles without radiation ¹¹	OTHER RECOMMENDED REGIMENS Taxane and cisplatin Paclitaxel 60 mg/m ² IV on Days 1, 8, 15, and 22 Cisplatin 75 mg/m ² IV on Day 1 Given for 1 cycle ¹² Docetaxel 60 mg/m ² IV on Days 1 and 22 Cisplatin 60–80 mg/m ² IV on Days 1 and 22 Given for 1 cycle ¹³ Docetaxel 20–30 mg/m ² IV on Day 1 Cisplatin 20–30 mg/m ² IV on Day 1 Weekly for 5 weeks ¹⁴ Irinotecan and cisplatin Irinotecan 65 mg/m ² IV on Days 1, 8, 22, and 29 Cisplatin 30 mg/m ² IV on Days 1, 8, 22, and 29 ⁶	OTHER RECOMMENDED REGIMENS—continued Paclitaxel and fluoropyrimidine Paclitaxel 45–50 mg/m ² IV on Day 1 weekly Fluorouracil 300 mg/m ² IV continuous infusion daily on Days 1–5 Weekly for 5 weeks ⁷ Paclitaxel 45–50 mg/m ² IV on Day 1 Capecitabine 625–825 mg/m ² PO BID on Days 1–5 Weekly for 5 weeks ⁷

^bLeucovorin is indicated with certain fluorouracil-based regimens. Depending on availability, these regimens may

#1: Paclitaxel + carboplatin (Ref: 2012 NEJM CROSS Trial)

Weekly Paclitaxel 50 mg/m² IV on Day 1 + Carboplatin AUC 2 IV on Day 1, and total 5 weeks.

#2: Fluorouracil + oxaliplatin (Ref: 2014 Lancet Oncol ACCORD 17)

Biweekly oxaliplatin 85 mg/m² IV on Day 1 + Leucovorin 400 mg/m² on Day 1 + Fluorouracil 400 mg/m² IV Push on Day 1 + Fluorouracil 800 mg/m² IV continuous infusion over 24 hours daily on Days 1 and 2, and cycled every 14 days for 3 cycles with radiation followed by 3 cycles without radiation.

#3: Fluorouracil + oxaliplatin (Ref: 2002 JCO Oxaliplatin in combination with protracted-infusion fluorouracil: report of a clinical trial for patients with esophageal cancer)

Oxaliplatin 85 mg/m² IV on Day 1, 15, and 29 for 3 doses + Fluorouracil 180 mg/m² IV daily on Days 1-33.

#11: Fluorouracil + cisplatin (Ref: 2002 JCO INT-0123/RTOG-9405)

Cisplatin 75-100 mg/m² IV on Days 1 + Fluorouracil 750-1000 mg/m² IV continuous infusion over 24 hours daily on Days 1-4, and cycled every 28 days for 2 cycles with radiation followed by 2 cycles without radiation (1st, 5th, 9th, and 13th week).

5. Cervical esophageal 怎麼做

Ans:

- (1) Definitive CCRT, 作法如上題所述
- (2) Elective nodal coverage 會包含 periesophageal, mediastinal, supraclavicular LNs region

6. Lower 1/3 延伸到 EC junction 2 公分以下怎麼做

Ans:

- (1) 團隊評估 neoadjuvant CCRT+OP or definitive CCRT
- (2) 依照 Siewert classification 區分 type I-III
- (3) 依照 esophageal cancer NCCN guideline, 若確認為 Siewert I & II 則依照食道癌治療, 因位置較低位 Distal third of esophagus and EGJ: elective nodal coverage 會包含 para-esophageal, lesser curvature, splenic nodes, and celiac axis nodal regions
- (4) 另參考 gastric cancer NCCN guideline:
With proximal gastric lesions or lesions at the EGJ: a 3- to 5-cm margin of distal esophagus and nodal areas at risk should be included. Nodal areas at risk include: perigastric, celiac, left gastric artery, splenic artery, splenic hilar, hepatic artery, and porta hepatic lymph nodes.

7. Bronchus 發現有 lesion, 外科說不開, 轉給你的話要怎麼處理

Ans:

- (1) Bronchus lesion 為 cT4b, 建議 definitive CCRT, 因較可能產生 esophageal perforation/fistula, 要密切注意相關症狀: substernal chest pain, high pulse rate, fever, pneumonia, cough, 視情況放 stent
- (2) 另外根據 NCCN 也可考慮 chemotherapy alone: Consider chemotherapy alone in the setting of invasion of trachea, great vessels, vertebral body, or heart

吳沅樺 (Rectal Ca)

1. Rectal cancer AAV 4cm, T3bN1M0 , 問 management 、 short course and long course 比較

Ans:

(1) Management 可選擇以下方式: (Ref: NCCN Version 1.2021)

- Neoadjuvant therapy
 - Long-course CRT (with concurrent Capecitabine or infusional 5FU) / short-course RT → Transabdominal resection → Adjuvant FOLFOX/CAPEOX
- Total neoadjuvant therapy
 - Long-course CRT (with concurrent Capecitabine or infusional 5FU) / short-course RT → Chemotherapy with FOFOX/CAPEOX (12-16 weeks) → Transabdominal resection
 - Chemotherapy with FOFOX/CAPEOX (12-16 weeks) → Long-course CRT (with concurrent Capecitabine or infusional 5FU) / short-course RT → Transabdominal resection

(2) Short course and long course 比較:

How does pre-op long-course chemoRT compare to short-course pre-op RT?

Bujko, Polish Study (Br J Surg 2006, PMID 16983741): PRT of 312 pts with cT3-4 with no evidence of sphincter involvement randomized to (a) 25 Gy/5 fx followed by TME within 7 days or (b) 50.4 Gy/28 fx with concurrent bolus 5-FU+LV followed by TME 4 to 6 weeks later. Primary endpoint was sphincter preservation. No difference in sphincter preservation, LR, OS, or DFS (Table 34.9). **Conclusion: Long-course chemoRT did not increase OS, LC, or late toxicity compared to short-course RT.** *Comment: Limitations to this study include clinical staging (no US or MRI), no standard post-op chemotherapy, not all TME, and no RT QA.*

	4-yr LR	4-yr DFS	5-yr OS	Grade 3-4 Early Toxicity	Grade 3-4 Late Toxicity	Positive CRM
Pre-op chemoRT	15.5%	55.6%	66%	18%	7%	4.4%
Pre-op short-course RT	10.6%	58.4%	67%	3%	10%	12.9%
p value	.2	NS	NS	<.001	.36	.017

Ngan, TROG Intergroup Trial (JCO 2012, PMID 23008301): PRT of 326 pts with cT3N0-2M0 rectal ACA, within 12 cm of verge (US or MRI staged) randomized to (1) 25 Gy/5 fx (given in 1 week), surgery in 3 to 7 days, six cycles 5-FU with folinic acid or (2) 50.4 Gy/28 fx + CONTINUOUS INFUSION 5-FU (225 mg/m²), surgery in 4 to 6 weeks, four cycles of 5-FU with folinic acid. Primary endpoint was LR. MFU 5.9 yrs. No differences in LR, DR, OS or late grade 3-4 toxicity (Table 34.10). For distally located tumors, LR was 12.5% in arm 2 versus 3% in arm 1, *p* = NS. **Conclusion: Short-course pre-op RT is equivalent to pre-op chemoRT without increased late toxicity. Unclear if short course is equivalent to long course for distally located tumors.**

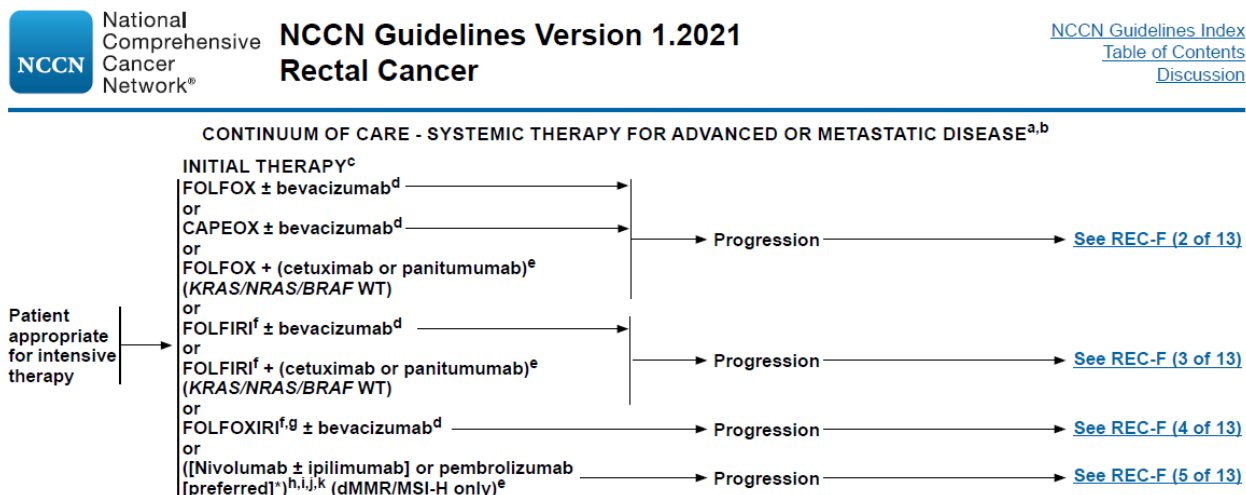
TROG 01.04	3-yr LR	5-yr DR	5-yr OS	Late Grade 3-4 Toxicity
Long course	4.4%	30%	70%	8.2%
Short course	7.5%	27%	74%	5.8%
p value	.24	.92	.62	NS

2. Rectal cancer, cT1N0, 切掉的話有甚麼 adverse features ?

Ans:

NCCN version 1.2021: High-risk features include positive margins, lymphovascular invasion, poorly differentiated tumors, or sm3 invasion (submucosal invasion to the lower third of the submucosal level).

3. MSI, KRAS, NRAS, BRAF 對於預後還有藥物選擇上有甚麼差異 ?



Ans:

(1) MSI: MSI-H 可使用 Nivolumab +/- ipilimumab or pembrolizumab (Ref: NCCN version 1.2021)

- Microsatellite instability (MSI), which results from MMR deficiency and is detected as changes in the length of repetitive DNA elements in tumor tissue caused by the insertion or deletion of repeated units (Ref: NCCN version 1.2021)
- Microsatellite instability have more favorable outcomes and lower likelihood of developing metastatic disease. (Ref: Perez P.1482)
- Pembrolizumab led to significantly longer progression-free survival than chemotherapy when received as first-line therapy for MSI-H–dMMR metastatic colorectal cancer. (Ref: N Engl J Med. 2020 Dec 3;383(23):2207-2218.)

(2) KRAS/NRAS/BRAF WT 可加入 cetuximab or panitumumab (Ref: NCCN version 1.2021)

- RAS alterations have a significantly shorter survival compared to wild type tumours.
- KRAS or NRAS mutations should not be treated with either cetuximab or panitumumab, either alone or in combination with other anticancer agents, because they have virtually no chance of benefit and the exposure to toxicity and expense cannot be justified. (Ref: NCCN version 1.2021)
- RAS wild-type (WT): the addition of Cetuximab to standard chemotherapy protocols such as FOLFOX and FOLFIRI resulted in improved prognosis.
- Patients with metastatic colorectal tumors carrying a BRAF mutation have a worse prognosis than those with the wild-type gene (Ref: CRYSTAL trial)

- BRAF V600E mutation makes response to panitumumab or cetuximab highly unlikely. (Ref: NCCN version 1.2021)

4. Anal cancer CCRT 的 indication ?

Ans:

- (1) RT is indicated in all cases except T1N0 tumors of anal margin treated with WLE, though RT alone may be appropriate for T1N0 (controversial).
- (2) For others (cT2-4 or N+), organ-preservation therapy is standard with concurrent chemoRT.

5. 承上題 CTV Dose, coverage ?

Ans:

(1) Sequential RT (Ref: NCCN 1.2021)

- Low risk elective nodal PTV volume: 30.6 Gy in 1.8 Gy daily fractions.
 - Low-risk elective nodal volume: gross disease CTV, high-risk elective nodal CTV, and presacral, bilateral internal, and external iliac nodes above the inferior border of the sacroiliac joint to the bifurcation of the internal and external iliac vessels at approximately L5/S1 vertebral body junction.
- High-risk elective nodal PTV is sequentially prescribed an additional 14.4 Gy in 1.8 Gy daily fractions for a total dose of 45 Gy.
 - The high-risk elective nodal volume: gross disease CTV plus the entire mesorectum, presacral nodes, and bilateral internal and external iliac lymph nodes inferior to the sacroiliac joint.
 - In patients with gross inguinal nodal involvement, the bilateral or unilateral inguinal nodes may be included in the high-risk elective nodal volume.
 - If there is no obvious involvement of the bilateral inguinal nodes, these are included in the low-risk elective nodal volume.
- Finally, for T1–2 lesions with residual disease after 45 Gy, T3–4 lesions, or N1 lesions, an additional 5.4–14.4 Gy in 1.8–2 Gy daily fractions is again sequentially prescribed to the gross disease PTV volume (total dose 50.4–59.4 Gy).

(2) RTOG 0529 (SIB)

Table 1: Dose Specification of Primary and Nodal Planning Target Volumes: RTOG-0529⁴

TNM Stage	Primary Tumor PTV Dose	Nodal PTV Dose
T2, N0	50.4 Gy (28 fxs at 1.8 Gy/fx)	42 Gy (28 fxs at 1.5 Gy/fx)
T3–4, N0	54 Gy (30 fxs at 1.8 Gy/fx)	45 Gy (30 fxs at 1.5 Gy/fx)
T any, N+ (≤3 cm)	54 Gy (30 fxs at 1.8 Gy/fx)	50.4 Gy (30 fxs at 1.68 Gy/fx)
T any, N+ (>3 cm)	54 Gy (30 fxs at 1.8 Gy/fx)	54 Gy (30 fxs at 1.8 Gy/fx)

6. Concurrent C/T regimen? 有沒有 trial 在討論不同 regimen? 為何不用 5FU+cisplatin

Ans:

(1) Concurrent C/T regimen



National
Comprehensive
Cancer
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NCCN Guidelines Version 1.2021 Anal Carcinoma

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PRINCIPLES OF SYSTEMIC THERAPY				
Localized Cancer		Metastatic Cancer		Subsequent Therapy
Preferred Regimens	Other Recommended Regimens	Preferred Regimens	Other Recommended Regimens	Preferred Regimens
• 5-FU + mitomycin + RT • Capecitabine + mitomycin + RT	• 5-FU + cisplatin + RT	• Carboplatin + paclitaxel	• 5-FU + cisplatin • FOLFECIS • mFOLFOX6 • Modified DCF	• Nivolumab • Pembrolizumab if not previously received

(2) 有沒有 trial 在討論不同 regimen? 為何不用 5FU+cisplatin?

Can MMC be replaced with cisplatin?

RTOG 98-11 showed that replacing MMC with cisplatin and addition of induction CHT significantly decreased hematologic toxicities but also increased colostomy rates and reduced disease-free survival and overall survival. However, ACT II trial established that replacement of MMC with cisplatin does not affect rate of complete response. Because of ACT II showing that CHT regimens are equal in terms of response rate, some have suggested results of 98-11 show that induction CHT may be detrimental.

Ajani, RTOG 9811 (JAMA 2008, PMID 18430910; Update Gunderson JCO 2012, PMID 23150707): PRT of 649 pts with T2-T4, N0-N3 disease randomized to (a) RT + concurrent 5-FU/MMC or (b) RT + induction/concurrent 5-FU/cisplatin. RT was 45 Gy/25 fx with 2 Gy/fx boost to primary tumor and involved nodes to between 55 and 59 Gy for those with T3-T4, N+ or T2 pts with residual disease after 45 Gy. Elective nodal sites were treated to 30.6–36 Gy/17-20 fx. CHT in standard arm was 5-FU 1,000 mg/m² continuous infusion days 1 to 4 and 29 to 32 and MMC 10 mg/m² IV bolus days 1 and 29. In experimental arm, two cycles of induction 5-FU were given and RT started on day 57 with third cycle of 5-FU and was continued through end of fourth cycle. Cisplatin was also given in four bolus administrations: 75 mg/m² IV days 1, 29, 57, and 85, with two cycles prior to start of RT. See Table 35.5. OS and DFS were superior in standard arm. **Conclusion: Use of concurrent 5-FU/MMC without induction CHT should remain standard of care.**

TABLE 35.5: Results of RTOG 98-11					
	5-yr OS	5-yr CFS	5-yr DFS	Grade 3-4 Heme Toxicity	Grade 3-4 Nonheme Toxicity
RT+ concurrent 5-FU/MMC	78.3%	71.9%	67.8%	61%	74%
RT+ induction/concurrent 5-FU/cisplatin	70.7%	65%	57.8%	42%	74%
p value	.026	.05	.006	.0013	NS

James, UK ACT II (Lancet Oncol 2013, PMID 23578724): PRT of 940 T1-T4 pts randomized in 2 x 2 fashion to either concurrent 5-FU/cisplatin or concurrent 5-FU/MMC followed by second randomization to 5-FU x two cycles after completion of chemoRT (maintenance) or observation. RT dose was 50.4 Gy/28 fx. MFU 5.1 years. CR was ~90% with either concurrent CHT regimen. 3-yr PFS was 74% with maintenance CHT versus 73% with observation (p = NS). **Conclusion: MMC/5-FU+RT remains standard. There was no benefit to addition of maintenance CHT after completion of standard therapy.**